

Lesson 19: Quasi-Linear First-Order PDEs - Lagrange's Method

Professor Dave Explains Differential Equations Master Series · Lesson 19

Video: Quasi-Linear First-Order Partial Differential Equations: Lagrange's Method Course: Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that a first-order PDE $Pu_x + Qu_y = R$ represents a geometric condition: the normal vector to a 2D surface $u(x, y)$ must be perpendicular to a direction field vector (P, Q, R) . Joseph-Louis Lagrange realized that solving the PDE is equivalent to integrating the streamlines of this vector field, known as Characteristic Curves. By solving a system of ordinary differential equations $dx/P = dy/Q = du/R$, we construct characteristic tracks along which the PDE reduces to simple calculus!

2. Core Theoretical Framework & Rigorous Formulations

1. The Lagrange-Charpit Characteristic System

For the quasi-linear PDE $P(x, y, u)\frac{\partial u}{\partial x} + Q(x, y, u)\frac{\partial u}{\partial y} = R(x, y, u)$:

$$\frac{dx}{P(x, y, u)} = \frac{dy}{Q(x, y, u)} = \frac{du}{R(x, y, u)}$$

2. Constructing the General Solution

1. Find two independent first integrals (conservation constants):

$$u_1(x, y, u) = c_1, \quad u_2(x, y, u) = c_2$$

2. The general solution is an arbitrary functional relationship:

$$F(u_1, u_2) = 0 \iff u_1 = \Phi(u_2)$$

3. To satisfy a Cauchy initial curve $u(x, 0) = f(x)$, use the initial condition to determine the specific functional form of Φ .

3. Fully Solved Whiteboard Problems

PDE Problem: Solving a Transport Conservation Law via Characteristics

Problem Statement: Solve the quasi-linear PDE $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2u$ subject to the Cauchy boundary condition $u(x, 1) = x^3$ for $x > 0$.

Step 1: Set Up Characteristic Ratios

$$\frac{dx}{x} = \frac{dy}{y} = \frac{du}{2u}.$$

Step 2: Solve First Characteristic Equation

$$\frac{dx}{x} = \frac{dy}{y} \implies \ln|x| = \ln|y| + C'_1 \implies \frac{x}{y} = c_1.$$

Step 3: Solve Second Characteristic Equation

$$\frac{dy}{y} = \frac{du}{2u} \implies 2\ln|y| = \ln|u| + C'_2 \implies \frac{u}{y^2} = c_2.$$

Step 4: Formulate General Implicit Solution

$$\frac{u}{y^2} = \Phi\left(\frac{x}{y}\right) \implies u(x, y) = y^2\Phi\left(\frac{x}{y}\right).$$

Step 5: Enforce Cauchy Boundary Condition $u(x, 1) = x^3$

Substitute $y = 1$: $u(x, 1) = 1^2\Phi\left(\frac{x}{1}\right) = \Phi(x)$. We are given $u(x, 1) = x^3$, so $\Phi(x) = x^3$ identically.

Step 6: Assemble Explicit Unique Solution

$$u(x, y) = y^2\Phi\left(\frac{x}{y}\right) = y^2\left(\frac{x}{y}\right)^3 = y^2\frac{x^3}{y^3} = \frac{x^3}{y}. \text{ Check: } xu_x + yu_y = x(3x^2/y) + y(-x^3/y^2) = \frac{3x^3}{y} - \frac{x^3}{y} = \frac{2x^3}{y} = 2u. \text{ Verified!}$$

⚠ Common Exam Pitfall & Trapdoor Avoidance

When determining $\Phi(x)$ from the boundary curve, replace the entire dummy argument $w = x/y$ into $\Phi(w)$. Do not mistakenly replace x inside the final answer before establishing the functional identity.

4. Key Takeaways & Exam Strategy Formula Card

Core Takeaway & Exam Rule

Lagrange's method maps 1st-order PDEs into ODE characteristic tracks $dx/P = dy/Q = du/R$.

- Video reference: [Quasi-Linear First-Order Partial Differential Equations: Lagrange's Method](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.